

A RANK-ZERO COUNTEREXAMPLE TO CONJECTURE 8.4 OF ABDI–CORNUÉJOLS–ZLATIN

ABSTRACT. We disprove Conjecture 8.4 of Abdi, Cornuéjols, and Zlatin as stated. Orienting the eight-vertex planar cubic bipartite multigraph 1110 of Exoo and Harary from one bipartition class to the other produces a planar sink-regular $(3, 4)$ -bipartite digraph D with $\rho(3, D) = 0$. The matroid $M_1(D, \mathbf{1})$ has unique basis $\emptyset$, while every rounded 1-factor has disconnected complement.

Conjecture 8.4 of Abdi, Cornuéjols, and Zlatin asserts that, for a planar sink-regular $(3, 4)$ -bipartite digraph $D = (V, A)$ and disjoint bases Q_1, Q_2, Q_3 of $M_1(D, \mathbf{1})$, there is a rounded 1-factor J such that $\text{dc}(J) = Q_1$ and $D \setminus J$ is connected [1, Conjecture 8.4]. Their conventions permit parallel arcs, and planarity and connectedness refer to the underlying undirected multigraph [1, pp. 2418, 2427].

Theorem 1. *Conjecture 8.4 of [1] is false, already for an eight-vertex planar instance with $\rho(3, D) = 0$.*

Proof. Let

$$S = \{u, b_1, b_2, b_3\}, \quad T = \{v, a_1, a_2, a_3\}.$$

For each $i \in \{1, 2, 3\}$, add one edge ua_i , two parallel edges between a_i and b_i , and one edge b_iv . This is the multigraph 1110 in [3, Figure 3]. Drawing its three u – v strands in parallel shows that the resulting graph G is planar; it is also connected, cubic, and bipartite.

Orient every edge from S to T , obtaining D . Let $\delta_D^+(U)$ be a dicut, and put $X = U \cap S$ and $Y = U \cap T$. Since $\delta_D^-(U) = \emptyset$,

$$|\delta_D^+(U)| = \sum_{z \in U} (d_D^+(z) - d_D^-(z)) = 3|X| - 3|Y|.$$

The underlying graph is connected and U is nonempty and proper, so the cut is nonempty. Thus every dicut has size at least 3. Since all vertices have degree 3, the digraph D is planar and sink-regular $(3, 4)$ -bipartite.

There are no active vertices, and

$$\text{disc}(V) = |T| - |S| = 0, \quad \rho(3, D) = 0$$

by [1, Lemma 3.7(1)]. Moreover, for every dicut shore U ,

$$\text{disc}(U) = |Y| - |X| = -\frac{1}{3}|\delta_D^+(U)| \leq -1.$$

2020 *Mathematics Subject Classification.* 05C20, 05C70.

Key words and phrases. dicut, dijoin, rounded 1-factor, cubic bipartite multigraph, Hamiltonian cycle.

By [1, Definition 4.7], $\emptyset$ is therefore the unique basis of $M_1(D, \mathbf{1})$: it has cardinality $\text{disc}(V) = 0$ and satisfies

$$0 = |\emptyset \cap U| \geq 1 + \text{disc}(U)$$

for every dicut shore. Set

$$Q_1 = Q_2 = Q_3 = \emptyset.$$

Each Q_i is a basis, and $Q_i \cap Q_j = \emptyset$ whenever $i \neq j$.

Let J be a rounded 1-factor. Since every vertex has degree 3, Definition 3.3 of [1] gives $|J \cap \delta_D(z)| = 1$ at every vertex z . Hence J is a perfect matching and $\text{dc}(J) = \emptyset = Q_1$.

There is a unique i for which $ua_i \in J$. The vertex a_i is then matched to u , so neither parallel edge between a_i and b_i belongs to J . Consequently $b_iv \in J$. After deleting J , the two parallel edges between a_i and b_i form a connected component of the underlying graph of $D \setminus J$. Hence $D \setminus J$ is disconnected. No rounded 1-factor satisfies the conjectured conclusion. $\square$

Remark 2 (Scope). The example lies in the rank-zero case and is not 3-connected: removing u and v leaves the three components $\{a_i, b_i\}$. Thus it does not address a positive-rank reformulation of Conjecture 8.4 or Barnette's conjecture. Nor does it disprove the planar three-dijoin statement in [1, Theorem 8.5]; it only falsifies Conjecture 8.4 as published.

More generally, the dicut and matroid calculations above apply to any connected cubic bipartite graph oriented from one bipartition class to the other, and a connected complement of a rounded 1-factor would be a Hamiltonian circuit. Thus simplicity alone does not repair the statement: the 26-vertex simple planar cubic bipartite non-Hamiltonian graph of Asano, Saito, Exoo, and Harary gives a simple counterexample [2].

REFERENCES

- [1] A. Abdi, G. Cornuéjols, and M. Zlatin, *On packing dijoins in digraphs and weighted digraphs*, SIAM J. Discrete Math. **37** (2023), no. 4, 2417–2461, doi:10.1137/22M1506511.
- [2] T. Asano, N. Saito, G. Exoo, and F. Harary, *The smallest 2-connected cubic bipartite planar nonhamiltonian graph*, Discrete Math. **38** (1982), no. 1, 1–6, doi:10.1016/0012-365X(82)90162-5.
- [3] G. Exoo and F. Harary, *The smallest cubic multigraphs with prescribed bipartite, block, Hamiltonian and planar properties*, Indian J. Pure Appl. Math. **13** (1982), no. 3, 309–312.